Mathematics Anxiety: It's Impact on Tomas Cabili National High School Grade 10-Acacian Student's Academic Performance and Well-being

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Abstract

This study examines the impact of Mathematics Anxiety: Its Impact on Tomas Cabili National High School Grade 10-Acacian Student's Performance and Well-being using a descriptive research design. The Mathematics Anxiety Rating Scale (MARS) was used to measure the anxiety levels of the participants. The results show that the students were moderately anxious when dealing with mathematical tasks. Despite experiencing moderate anxiety, the students' academic performance, assessed through the grading system, indicated that all students passed, with the majority achieving a Very Satisfactory rating. The data were analyzed using Frequency count, Percentage, Mathematics Anxiety Rating Scale (MARS), and the Grading System to ensure accuracy and reliability. The findings suggest that while moderate mathematics anxiety was present, it did not significantly hinder the students' academic success. The study recommends providing continued support and implementing strategies to reduce anxiety and enhance both academic performance and emotional well-being.

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Dedication

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Table of Contents Page

Title Page i

Approval Sheet ii

Abstract iii

Acknowledgement iv

Dedication v

Table of Contents vi

List of Tables viii

List of Figures ix

Chapter

1 THE PROBLEM AND ITS SETTING

Introduction 1

Theoretical Framework 5

Conceptual Framework 7

Statement of the Problem 7

Null Hypothesis 10

Significance of the Study 10

Scope and Delimitation 12

Definition of terms 12

2 REVIEW OF RELATED LITERATURE AND STUDIES 15

3 METHODOLOGY.....27

Research Design 27

Research Locale 27

The Respondents.....27

Samples and Sampling Techniques 29

Research Instrument and Validation of Instrument 29

Data Gathering Procedure.....29

Statistical Treatment.....30

4 PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA 29

5 SUMMARY, CONCLUSION AND RECOMMENDATIONS 37

REFERENCE LIST.....40

APPENDICES 52

Appendix A Letter to the School Principal 52

Appendix B Letter to the Respondents.....53

Appendix C Documentation.....54

Appendix D Survey Questionnaire 55

CURRICULUM VITAE 70

List of Tables **Page**

Table 1 Factors contributing mathematic anxiety that impact students performance.....32

Table 2 Anxiety level of the Grade 10-Acacian students of Tomas Cabili National High School.....33

Table 3 Mathematics anxiety affects students academic performance in the subject.....34

Table 4 Strategies used to manage mathematics anxiety 35

Table 5 Current mathematics grade range of Grade10-Acacian students.....36

List of Figures Page

Figure 1 Research Paradigm of the Study 8

Figure 2 Map of Tomas Cabili National High School 28

Figure 3 Documentation 54

CHAPTER I

PROBLEM AND ITS SETTING

Introduction

Everyone experienced anxiety occasionally, but people who suffered from anxiety disorders frequently experienced extreme and overwhelming concerned and feared. These feelings were typically accompanied by a variety of behavioral and cognitive symptoms as well as physical discomfort. If left untreated, they could cause significant distress, was difficult to managed, and last for a long period. Anxiety disorders could have interfered with day to-day activities and had a negative impact on a person's social interactions, family, and performance at job or school (World Health Organization: WHO, 2023).

Math anxiety was the feeling of tension and apprehension that interfered with math's performance ability, the manipulation of numbers and the solving of mathematical problems in a wide variety of ordinary life and academic situations (Khasawneh et al., 2021). They added that self-awareness contributes to math confidence, math-related values, and self-efficacy. By improving these on influences, math anxiety may be overcome and performance could improve.

According to Brooks, (2020), tests and classroom settings were not the only places where mathematics anxiety occurs. It could also have applied to a variety of real-world circumstances, making otherwise capable and competent people kept clear of any mathematically related issues or even occupations that call for mathematical expertise. It's crucial to remember that mathematics anxiety was different from general assessment anxiety; it relates directly to anxiety over mathematics rather than test anxiety. It was distinct since there were no common anxiety problems associated with other particular courses, such as history, writing, or reading.

As stated by Sara, (2024), many of students claimed to disliked mathematics. However, for certain individuals, the concerned was more than just a disliked of fractions or algebra. Some students experienced negative emotions, such a nervousness of failing, while they worked with math. This might have made it more difficult for them to performed well.

Furthermore, students with math anxiety might experience disruptions in their working memory as early as the first grade. Working memory served as a short-term memory to assisted us remember numbers. According to study of Cherry, (2023), dealing with short-term memory loss could be difficult and upsetting because short-term memory was essential to our daily life. Unfortunately, Math anxiety could cause both primary and high school learners to lose this memory. Thus, Math anxiety could cause students to fall up to six months behind their peers in the subject. Moreover, it was important to developed productive studied habits to helped people who did not experienced mathematics anxiety when math ideas got more difficult.

Mathematics was fundamental to our civilization. In many of our daily tasks, mathematics was used. It had a major impact on how people thought and reasoned. Strong mathematical abilities, and mathematical techniques to carry out solutions. It improves our capacity for critical thought when confronted with difficult problems and frequently motivates us to reflect and go deeper. We must come up with solutions for the majority of the choices, tactics, disputes, and difficulties we faced in life. By encouraging mental flexibility and the knowledge that there was frequently only one right answered to every issued, mathematics helped us to investigate and made well-informed decisions (Oswal, 2022).

Moreover, Mathematics was a subject that many students struggled with at some time in their learning. Thus, Math anxiety was more than just a disliked of the topic for people who suffered from it. It could cause strong emotions of fear and failure that impaired performance. When these students had to dealt with math difficulties, the pressure and lack of confidence they felt might cause their minds to blank out, causing them to forgot even the concepts they understand. Math difficulties were not usually associated with a learning limitation. Lack of a strong foundation for success is the problem for many learners. Their grades might had declined as a result of decreasing behind in a particular unit or moving on to more difficult material before they were competent. Some students could discover struggle to learn math, particularly if they did not possess the resources or support they require. Students shouldn't have an unfortunate encounter with math considering concerns such math anxiety and learning difficulties (Stanley, 2021).

Math anxiety was a common problem. This made someone anxious when given the opportunity to do math problems, which impairs their ability to thought effectively. Academic challenges could have developed from an ongoing cycle of low self-esteem and apprehension (West,2022). Moreover, for many students, mathematics had been perceived as a challenging subject, which usually leads to emotions of fear and detachment. Since learning anxiety was so common, it coud hinder students' comprehension of mathematical ideas and had an impact on their overall academic performance. However, identifying the motivating factors that support learners in succeeding in mathematics could enhance both their performance and learning ability (Supriadi et al., 2024).

According to the study of Lockett, (2022), even easy math problems might cause a variety of feelings and symptoms in a person who suffered from math anxiety. Moreover, Emotional symptoms include increased anxiety, tension, trepidation, and a feeling of impending doom. Physiological symptoms include sweating, lightheadedness, and a beating heart. In more extreme circumstances, math-related circumstances might even trigger a panic attack. Cognitive symptoms having trouble focusing and using working memory, not only in arithmetic but also in other subjects like reading. Furthermore, Mathematics anxiety could eventually have led to poor performance at work and in school, which could result in problems like poor performance reviews and low grades. This could have led to feelings of low self-esteem, decreased confidence, generalized anxiety, and more if left untreated.

In the Philippines, there were a number of challenges with math education. Low performance and underachievement in general. Furthermore, the average of mathematics scores of Filipino students did not significantly improve, as anticipated by the Department of Education (DepEd), and they remained low when compared to students from other nations who participated in the most recent Programme for International Student Assessment (Pisa) (Bautista, 2023). The PISA, developed by the (Organization for Economic Cooperation and Development) OECD, measures 15-year-olds' ability to use their reading, mathematics and science knowledge and skills to meet real-life challenges. The 2022 PISA results, the Philippines ranked sixth from the bottom in mathematics with an average score of 355 (Montemayor, 2023).

According to Servallos, (2023), the Vice President and Education Secretary Sara Duterte admitted that the 2022 PISA results "bear uncomfortable truth," but it

should be a jump-off point for the country's education system to work on its strengths and weaknesses. "The PISA results are not merely a reflection of our education system. It is a mirror reflecting our collective efforts, investments and most importantly, our commitment to education and the future we envision for our children," Duterte said in a statement. "As such, this is a call to action, a call to our collective responsibility as a nation. I call on everyone to pull our efforts together for a more resilient MATATAG education system, an education system that aims to improve learning outcomes, prioritize student and teacher well-being, and promote accountability to close remaining disparities," she added.

The research focused on how mathematics anxiety had an impact of Tomas Cabili National High School Grade 10 student's performance and well-being. The purpose of the research was to present information for reflecting and planning how mathematics anxiety can be reduced and regulated in classrooms. Additionally, this study seeks to provide valuable insights for educators to developed effective strategies to support students struggling with mathematics anxieties.

Theoretical Framework

This study was anchored on the theories of Control-Value Theory of Achievement Emotions by Reinhard Pekrun 2014, Self-Determination Theory by Richard Ryan and Edward Deci 2022 and Deficit Theory by Sheila Tobias 1986.

Control-Value Theory of Achievement Emotions by Reinhard Pekrun 2014. This theory explored the impact of emotions on academic performance and engagement. It suggests that an individual's feelings regarding their sense of control and the values they attached to their achievements play a crucial role in shaping their emotional responses to success and failure.

Self-Determination Theory (SDT) by Richard Ryan and Edward Deci 2022. This emphasis that mathematics anxiety could be primarily attributed to an unfulfilled sense of autonomy, competence, and relatedness. This lack of fulfillment could create feelings of being controlled or inadequate when faced with math problems, resulting in anxiety. Essentially, when students felt they lack agency in their learning or lack confidence in their math skills, they were more prone to experiencing anxiety in math-related situations.

Deficit Theory by Sheila Tobias 1986.Emphasis struggling with math could increase anxiety levels, which may result in a decreased interest in studying the subject. According to this theory, individuals with weaker math skills were more prone to experiencing math anxiety (Zhang et al., 2019).

In relation to this theories, the present study aimed to analyzed how mathematics anxiety affect the academic performance of the learners. Thus, the fear of math could trigger physical and emotional stress during math-related tasks and impacting overall well-being.

Conceptual Framework

As this study was anchored from the different theories mentioned above, the study arises in concept and variables that need to be examined, and with that, this framework was conceptualized.

The first frame enumerated the independent variables. These were the identified as respondents profile in terms of: age sex cultural background education background and mathematics anxiety.

For the intervening variables presented in the second frame, the researchers included the Environmental Factors: Classroom dynamics, teaching methods, and peer interactions. Intellectual Factors: Cognitive abilities, prior knowledge, and learning styles. Personal Factors: Self-esteem, motivation, and personal attitudes towards mathematics.

For the dependent variable presented in the last frame, the researchers included the Academic Performance: grades, test scores, achievement in mathematics and Overall Well-being: mental health, emotional stability, social interactions.

Figure 1 showed the interplay of independent variable, intervening variable and dependent variables as analyzed and used to create specific strategies to reduce mathematics anxiety. Ultimately improving both the academic performance and overall well-being of Grade 10-Acacian students.

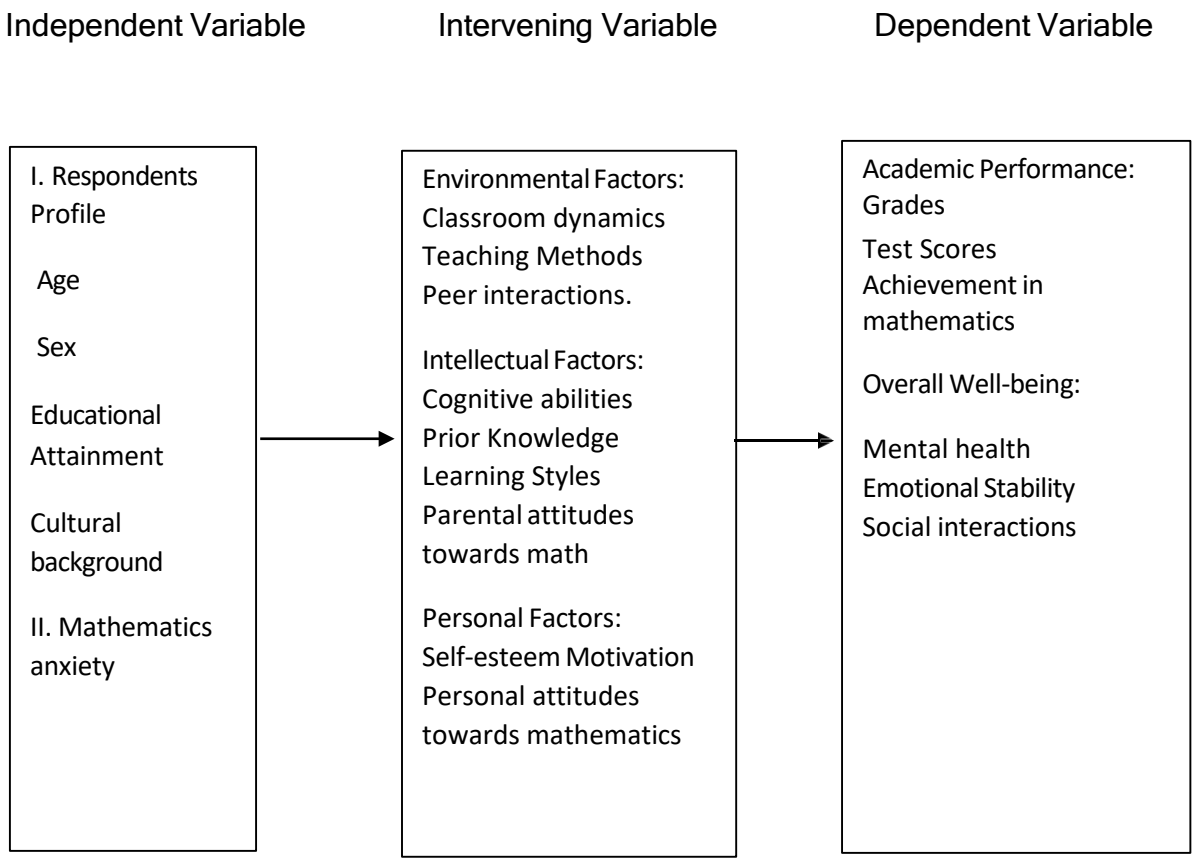


Figure1. Research Paradigm of the Study

Statement of the Problem

This study aimed to determine the Mathematics Anxiety Its Impact on Tomas Cabili National High School Grade 10-Acacian Student's Performance and Well-being.

Specifically, this study tried to answered the following:

1. What were the factors contributing to mathematics anxiety that impact students performance?
2. What was the anxiety level of Grade 10-Acacian students of Tomas Cabili National High School?
3. How did mathematic anxiety affect students academic performance in the subject?
4. What were the strategies do Grade 10-Acacian students use to manage their mathematics anxiety?
5. What was the current mathematics grade range of Grade 10-Acacian students?

Null Hypothesis

From the foregoing statement of the problem, the following no hypothesis were tested:

HO1: There was no significant difference on the factors contributing to mathematics anxiety that impact students performance.

HO2: There was no significant difference on the anxiety level of Grade 10-Acacian students of Tomas Cabili National High School.

HO3: There was no significant difference on mathematics anxiety affect students academic performance in the subject.

HO4: There was no significant difference on the strategies do Grade 10-Acacian students use to manage their mathematics anxiety.

HO5: There was no significant difference of the current mathematics grade range of Grade 10-Acacian student.

Significance of the Study

The findings of this study assists learners, teachers, parents, and future researchers at Tomas Cabili National High School in developing a deeper understanding of math anxiety.

The result of the study proved beneficial to the following:

The Students. Not every student enjoys mathematics. In fact, many learners experience tension, apprehension, and fear when faced with mathematical tasks in class, which negatively impacts their performance in math-related activities. This study

would be beneficial for these learners, helping them recognize that they might had anxiety issues related to mathematics.

The Teachers. The data provided would serve as a guide for teachers in addressing the needs of students who experience math anxiety. The results would offer valuable insights into students' attitudes toward math anxiety. The study identifies factors that influence the anxiety levels of junior high school students, particularly those in Grade 10. Consequently, teachers could support their students in developing a more positive attitude toward mathematics.

The School Administrators. This study would enable administrators to assess the level of anxiety among Grade 10 learners in mathematics. The data gathered would inform the Mathematics policy that each school promotes or advocates, allowing for transformation and improvement that benefits the entire school community, based on the quantitative evidence presented in the study.

The Parents. Similar to teachers, parents would gain crucial insights into math anxiety and be made aware that their children might be experiencing it. This understanding would help parents empathize with their children's challenges.

The Future Respondents. Future researchers would find valuable ideas and insights regarding the math anxiety levels of Grade 10 learners in this study. This research could help reduce assumptions and speculations among readers. The outcomes of this study would be advantageous for both current and future research endeavors, potentially serving as a foundation for new theories in learning.

Scope and Delimitation

This study examined the mathematics anxiety experienced by Grade 10-Acacian students at Tomas Cabili National High School in Iligan City during the 2024-2025 school year. A total of 30 Grade 10-Acacian students participated in the research.

The focus was on assessing the level of mathematics anxiety among these learners using the Mathematics Anxiety Rating Scale (MARS), which categorizes anxiety into mild, moderate, high, and very high levels. Additionally, the study aimed to evaluate the academic performance of the respondents and to explore between mathematics anxiety and their academic performance.

Definition of terms

To assist better understanding of the readers the following terms were defined conjecturally and operationally to further explicate them as to how they appeared in the paper.

Anxiety. This referred to a normal emotional reaction marked by anxiety, fear, or worry over what might happen in the future. While a certain amount of anxiety might spur activity, too much anxiety could impair wellbeing and performance by making daily chores seem burdensome.

Mathematics. It was defined as a discipline that deals with amounts, forms, patterns, and numbers. Mathematics was a method of critical thinking and problem-solving that was useful in many facets of life, from engineering to budgeting. It was not only about arithmetic.

Math anxiety. This term referred to a particular kind of anxiety that people experience when they have to complete math-related tasks. It might show up as panic, anxiety, or terror and frequently results in a complete avoidance of math.

Knowledge. This referred to an information, comprehension, and skills gained via experience or education. The basis for problem-solving and critical thinking in mathematics was knowledge, which ranges from fundamental arithmetic to complex ideas.

Self-confidence. This referred to the general feeling of value or self-worth that people have. While low self-esteem can result in feelings of inadequacy and increased anxiety, especially in academic contexts, high self-esteem could enable people to take on difficulties, including those in mathematics.

Short-term memory. It was defined as the ability to temporarily store and work with a small amount of data. Because it enables people to mentally track numbers and steps, short-term memory was essential for mathematical computations and problem-solving.

Well-Being. This denoted to a comprehensive condition of health that takes into account social, psychological, and emotional facets of existence. A number of things affect well-being, such as scholastic achievement and personal fulfilment, and stressors like math anxiety can have a harmful effect.

Mathematical ability. This referred to the ability to comprehend, analyse, and work with mathematical ideas and procedures. Academic achievement is greatly impacted by this skill, which differs from person to person and can be modified by things like practice, education, and confidence.

Mathematical technique. This referred to the particular techniques or approaches applied to mathematical issues. Gaining proficiency in these methods can improve a person's confidence and mathematical ability. They could range from simple algorithms to sophisticated approaches to problem-solving.

Academic performance. It was defined as Grades, test scores, and general comprehension of the subject matter were frequently used to gauge a student's level of success in their academic endeavours. Knowledge, self-esteem, and anxiety levels were some of the variables that might affect academic achievement, especially in disciplines like mathematics.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter entailed the discussion and on the related literature and studies which were significant and relevant on the analysis and interpretation of data of the study. It provided reliable information, theories, and ideas that show relationship to the study.

Related Literature

Math anxiety referred to the feelings of tension, helplessness, mental disorganization, and heightened physiological responses that people experience when faced with math (Luttenberger et al., 2018). Moreover, Math anxiety manifested as a state of unease triggered by engaging in math tasks, marked by feelings of apprehension, worry, aversion, and frustration. This condition was influenced by various factors, including math self-concept, teaching methods, and gender differences. According to the study of Van Mier et al., 2019, that males tend to perform better in math tasks compared to females while females generally experience higher levels of math anxiety than their male counterparts.

One of the academic areas that seems to make people more anxious than others was mathematics (Guo & Liao, 2022). As a fundamental subject in K-12 educations around the world. Mathematics helped students developed their cognitive skills, such as critical thinking, reasoning, and problem-solving abilities, which were vital for success in complex situations in the future (Donolato et al., 2020. It also gave them the knowledge and abilities they needed to navigate a changing world. Math anxiety was the subject of an increasing number of research (Moscoso et al., 2020).

Even after taking into consideration their general mental and mathematical skills. Students who suffered from mathematics anxiety generally perform worse in math operations than their classmates (Li et al., 2021). Even while the students understand that mathematics was an essential topic that would provide them with valuable abilities for their future jobs, they would also avoid activities involving the subject (Hasty et al., 2020).

Many have argued that learning and performance in mathematics were significantly influenced by a complex combination of cognitive, emotional, and environmental elements (Cipora et al., 2022). Due to its importance in both academic and everyday contexts, mental calculation had been a key area of research in the broad subject of mathematics (Xu et al., 2021). Additionally, several emotional characteristics, like math-related viewpoints and anxiety, were directly associated with mental calculation (Lichtenfeld et al., 2023).

The problem of math anxiety was complicated (Lukowski et al., 2019). Furthermore, he stated that, math anxiety could be divided into two or more interrelated categories: test anxiety and math learning anxiety. While math test anxiety is directly associated with the experience of taking exams, math learning anxiety was related to executing mathematical operations, managing numbers, or understanding math concepts in a classroom context (Luttenberger et al., 2018). Math performance could be severely impaired by both forms of math anxiety (Passolunghi et al., 2020). The various questionnaires used to measure math anxiety reflect its complexity, such as the Mathematics Anxiety Questionnaire (MAQ), the Mathematics Anxiety Rating Scale (MARS), the Mathematics Anxiety Scale (MAS), and the Anxiety Toward Mathematics Scale (ATMS). The MAS and MAQ concentrate on the unfavorable emotions connected to mathematics. The ATMS assesses overall discomfort in math-related

circumstances, whereas the MARS was a comprehensive measure that focuses on anxiety connected to math tests and numerical tasks.

According to the study of Commodari & La Rosa, (2021), it's important to distinguish math anxiety from other forms of be concerned, such general academic anxiety. Regardless of the subject's uniqueness, general academic anxiety creates negative emotions that encompass all academic settings. Additionally, it might be connected to certain circumstances that are thought to pose a risk to one's self-esteem, including important tests (standardized academic test anxiety). In addition to influencing academic and professional growth throughout one's life, general academic anxiety can lead to academic laziness, which is an intentional delay of crucial and planned academic-related acts or decisions. Sweaty hands, cold, anxiety, panic, rapid breathing, a racing heartbeat, and stomach ache were the primary physiological symptoms.

On the other hand, Paechter et al., (2017), psychological symptoms include anxiety before tutorial lessons, panic attacks, losing focus during tests, feeling powerless while completing assignments or homework, or a lack of interest in academic subjects. Math anxiety, as opposed to generic academic anxiety, was associated with particular difficulties in handling activities involving numbers or math. Math anxiety was characterized by low self-esteem and poor performance in math reasoning tasks, numeracy processing, and math-related evaluation scenarios. As previously mentioned, the two distinct components of math anxiety—math learning anxiety and math exam anxiety—can have distinct effects on calculating abilities and require more research. Several studies suggested that anxiety connected to other specialized courses might also exist, even though the majority of research has concentrated on math anxiety, thinking that math-related fears are more intense than

those experienced concerning other academic subjects (Dowker et al., 2016). In this context, a fascinating study by Punaro and Reeve (2012) demonstrated that while literacy anxiety was less common than math anxiety, it could nonetheless occured in kids with age-appropriate reading abilities. According to other research, reading difficulties and anxiety symptoms were related (Carroll & Iles, 2006), and children and adolescents with low literacy levels exhibit higher levels of linguistic anxiety than their literate peers.

Moreover, Math performance from various skills and knowledge related to using numbers to observed, compared, predicted, quantified, and verified relationships (Gersten et al., 2005). The primary skill that children developed at the beginning of primary school was calculation. McCloskey et al. (1985) proposed that two distinct systems were involved in calculation. The first, known as the “number-processing system,” encompasses various components that facilitate number understanding and production. The second, referred to as the “calculation system,” pertains to the mechanisms specifically engaged in calculation. This system includes three components: processing operational symbols or words, storing basic arithmetic facts, and executing calculation procedures. Successful completion of any calculation task requires the effective functioning of both systems. Additionally, mental and written calculations involve different procedures and processes (Cornoldi and Lucangeli 2004). According to McCloskey et al., (1985), identified four key components of calculation skills: numerical knowledge, calculation accuracy, calculation speed, and written calculation. Numerical knowledge encompasses the abilities and understanding that enable a child to grasp numerical measures and their transformations (Dehaene et al., 1990).

Calculation accuracy and speed relate to the ability to perform arithmetic calculations both accurately and quickly (Cornoldi & Lucangeli, 2004; Jordan & Montani, 1997). Written calculation involves applying the correct procedures during written tasks. Poor performance in mental and written calculation tasks can be attributed to difficulties in one or more of these components that made up calculation skills (Cornoldi & Lucangeli, 2004).

Research indicated that children generally had positive feelings about mathematics, but these could shift to negative attitudes over time (Dowker et al., 2012; Petronzi et al., 2018). Studies had shown that math anxiety can emerge as early as first grade and continue throughout elementary school (Ganley & McGraw, 2016 and Sorvo et al., 2017). This anxiety could hinder children's ability to learn math, as negative feelings often lead to avoidance and poor performance, which in turn intensifies their anxiety (Vukovic et al., 2013). Additionally, research had found that children with developmental dyscalculia were twice as likely to experience math anxiety compared to their peers who performed typically in mathematics (Devine et al., 2018).

Furthermore, several studies had found a moderate negative relationship between math anxiety and math performance (Justicia-Galiano et al., 2017; Passolunghi et al., 2020). While moderate levels of anxiety could actually motivate students to grasp mathematical concepts more effectively, high levels of math anxiety were linked to poor performance, avoidance behaviors, and negative self-beliefs regarding their skills (Ashcraft, 2016). A recent meta-analysis by Zhang et al. (2019) highlighted a strong negative correlation between math anxiety and performance, particularly among high school students compared to those in elementary school. However, the exact mechanisms behind this relationship remain

unclear. Research had indicated that working memory plays a crucial role in mediating the connection between math anxiety and performance in children (Justicia-Galiano et al., 2017). Thus, Ashcraft, (2016), noted that math anxiety particularly impacts calculations that demand high working memory. On the other hand, Maloney et al. (2012) argued that math anxiety influences all types of mathematical tasks, even the most fundamental ones like magnitude comparison. A recent study by Lee and Cho (2018) found a negative correlation between math anxiety and performance in tasks related to magnitude processing and complex calculations that require significant working memory. In contrast, simple calculations that rely on retrieval and impose a low working memory load are not significantly affected by math anxiety. Beilock and Willingham (2014) established a strong connection between math anxiety and performance, especially in mathematical reasoning and advanced problem-solving skills. Conversely, Harari et al., (2013) found a weak association between math anxiety and digital calculation, which evaluates basic computational abilities.

The evidence suggested that math anxiety had a significant negative effect on students' math performance (Van Mier et al., 2018; Wu et al., 2012; Zhang et al., 2019). However, a recent meta-analysis by Zhang et al. (2019) points out that previous studies examining the link between math anxiety and performance primarily focused on calculation and problem-solving skills, without exploring how math anxiety affects various components of math performance. Additionally, much of the existing literature had concentrated on math anxiety without differentiating it from general academic anxiety. Therefore, it was crucial to better understand the relationship between math anxiety and math performance, particularly in identifying which specific math skills were most adversely impacted by math anxiety.

For many children, mathematics had been perceived as a challenging subject, which could cause anxiety and disinterest. Students' overall academic performance may suffer because of this learning anxiety, which could hinder their comprehension of mathematical ideas (Eshet et al., 2021). Nonetheless, it might enhance students' learning experiences and results by investigating the motivating elements that propel them to excel in mathematics (Cho & Heron, 2015; Kim et al., 2014). Examining how learning anxiety, mathematical reasoning, and motivation in mathematics relate to one another across grade levels could reveal useful teaching tactics to address these problems.

Since the ability to create useful conclusions and comprehend cause-and-effect interactions depends on one's prior knowledge, mathematical reasoning was crucial (Dong et al., 2020). The ability to used reasoning to determine if a mathematical law was true or untrue was an essential part of this process (Dreher, 2020). This learning objective emphasizes how important it is for students to apply their reasoning skills when solving mathematical issues. Because it incorporates the creation of concepts and hypotheses required to overcome mathematical obstacles, advanced talent of reasoning was essential in problem-solving (Kurnaz, 2018). Math anxiety, mathematical reasoning and motivation to study are some of the characteristics that affect how students approach problem-solving (Zhou et al., 2020).

Research on math was learning anxiety highlights that the complexity of math problems can lead to feelings of apprehension, which in turn can result in mistakes during problem-solving. However, it's worth noting that the study focused on elementary and secondary-level mathematics teachers, leaving the implications for students largely unexplored. In a similar, found a small yet significant difference between student anxiety and their problem-solving abilities, specifically among

elementary school students. The Self-determination theory (SDT) proposed by Ryan & Deci (2022) emphasizes autonomy, competence, and relatedness as key factors that affect motivation and overall well-being. By examining variables that are relevant to primary school students, we can investigate how environments that foster autonomy, promote competence, and encourage positive relationships can enhance intrinsic motivation and psychological well-being. This indicates a need to explore this relationship among primary-level students (Ryan & Deci, 2022).

Previous studies had shown that learning anxiety could negatively affect student's motivation to learn (Jiang et al., (2022). He furthered highlighted the need to address learning anxiety in order to create a positive learning environment and boost overall student motivation. This indicates that gaining a deeper understanding of how learning anxiety interacts with cognitive processes like mathematical reasoning and problem solving could offer valuable insights into its impact on learning motivation. Additionally, research had found a significant positive correlation between anxiety and both intrinsic and extrinsic motivation, as well as a motivation (Luo et al., 2020). These findings provide theoretical backing for the potential used of anxiety to enhance motivation in foreign language learning settings. Moreover, studies have indicated that strong mathematical reasoning skills contribute to better problem-solving abilities. In research conducted by Liang & She (2021), students with higher mathematical reasoning skills showed improved performance in mathematical problem solving. This suggests that the link between mathematical reasoning and problem solving may be crucial in influencing students' motivation to learn. Furthermore, Román-González et al. (2016), observed a positive relationship between reasoning and problem-solving skills in primary and secondary education. The connection between mathematical problem solving and motivation has been emphasized in earlier studies.

Additionally, Schukajlow et al. (2022a), explored the interconnectedness of problem solving and motivation, indicating that positive experiences in problem solving can enhance students' motivation to tackle more mathematical challenges.

Essentially, motivation plays a crucial role in influencing mathematical problem-solving (Özcan, 2016). Additionally, Özcan & Eren Gümüş (2019) clarified that math anxiety had an indirect effect on mathematical problem solving by influencing self-efficacy, although they did not elaborate on how mathematical problem solving acts as a mediating factor. A study by Zhou et al. (2020), revealed that the relationship between teachers and students affects students' mathematical problem-solving skills through self-efficacy and math anxiety. However, the effect of self-efficacy on math anxiety is relatively minor, and the study did not clarify the direct link between anxiety and student problem solving, omitting the involvement of teacher subjects.

In addition to these factors, students' problem-solving skills could also be affected by their motivation to learn (Al-Kiyumi & Albeloushi, 2021; Chen & Lin, 2020). Learning motivation was fundamentally an internal drive that encourages individuals to pursue specific goals (Filgona et al., 2020). Several factors can influence students' motivation to learn, such as their aspirations, eagerness to learn, overall well-being, classroom atmosphere, engaging learning activities, and the effectiveness of teachers in presenting lessons (Komariah et al., 2022). Many studies have investigated the relationships between learning anxiety, mathematical reasoning, mathematical problem solving, and student motivation. However, there remains a gap in research concerning how improving mathematical reasoning affects other mathematical skills, like problem solving (Demir et al., 2023; Demir & Zengin, 2023; Geteregechi, 2023; Smit et al., 2022). Although individual aspects have been studied, the combined effects of learning anxiety, mathematical reasoning, and mathematical problem solving in predicting student learning motivation have not been thoroughly studied. By examining the complicated relationships between these factors and how they affect the development of an engaging learning environment in mathematics education, this study aims to close that gap.

Related Studies

Foreign

Previous studies had demonstrated that a number of characteristics, including working, math anxiety, and stereotypes, may influence the relationship between gender and mathematical ability (Barroso et al., 2020). However, the majority of research only incorporate one particular mathematics exam or concentrate on one particular predictor. Therefore, it was yet unknown how much math performance was influenced by working memory, math anxiety, and gender stereotypes taken together. Whether working memory, mathematics anxiety, and gender-related stereotypes play different roles in basic and elaborate mathematical assessments was yet unknown. In this study, that evaluates arithmetic fluency in addition to a more challenging mathematics work that requires the capacity to override basic answers and apply advanced problem-solving techniques (Peng et al., 2015).

Math proficiency was frequently associated with boys, whereas speaking abilities were associated with girls. Nonetheless, studies shown that men and women did not naturally differed cognitively in their abilities for mathematics, eliminating the myth of a male "math brain." Nevertheless, others have called this myth one of the most harmful myths in modern day. It was a fact that practically anyone could become highly proficient in math (The Myth of the Male Math Brain — AAUW: Empowering Women Since 1881, 2020).

Anxiety among students was acknowledged as a serious problem in Vietnam. 500 university and college students in Ho Chi Minh City participated in a study conducted by Thanh Tra (2019), which identified factors that negatively impact students' mental health and showed a significant prevalence of anxiousness.

Additionally, Van (2018), discovered that a large number of high school students in Ho Chi Minh City suffer from academic anxiety as a result of self-perceptions and cognitive distortions, issues with their relationships with teachers, and parental pressure surrounding academic achievement. The purpose of the current study was to investigate mathematics anxiety in Vietnamese high school students and investigate the relationship between academic coping mechanisms and mathematics anxiety.

Local

In the Philippines, there were a number of challenges with math education. Low performance and underachievement in general. Furthermore, the average of mathematics scores of Filipino students did not significantly improve, as anticipated by the Department of Education (DepEd), and they remained low when compared to students from other nations who participated in the most recent Programme for International Student Assessment (Pisa) (Bautista, 2023).

The PISA, developed by the (Organization for Economic Cooperation and Development) OECD, measures 15-year-olds' ability to use their reading, mathematics and science knowledge and skills to meet real-life challenges. The 2022 PISA results, the Philippines ranked sixth from the bottom in mathematics with an average score of 355 (Montemayor, 2023).

According to the study of Angeles et al., (2023), she conducted in Morong National High School with 253 respondents out of 829 10th Grade students. The probability sampling was employed through simple random sampling on 10th Grade students regardless of their age, gender, section, or academic achievement in Mathematics. According to the statistical analysis of the results, the Pearson

coefficient value suggests a weak negative correlation between math anxiety and mathematical performance, however, the magnitude of the correlation was quite small, indicating that there was only minimal relationship between the two variables. There was a weak negative correlation between math anxiety and mathematical performance of Grade 10 students in Morong National High School. Furthermore, the results of the scaling of math anxiety level resulted in quite a bit of anxiety on the respondent's mathematical performance. Many factors were also identified that influence the students' math anxiety: the level of difficulty of math lessons, level of difficulty of math problems, oral recitation, low self- esteem, lack of knowledge in math, and surprise quizzes.

Synthesis of the Reviewed Literature and Studies

Math anxiety was the tension, fear, or nervousness that people felt when dealing with mathematical problems. Thus, could greatly hinder a person's ability to tackled math tasks, resulting in lower performance and often causing them to avoid mathematics entirely. In essence, high levels of math anxiety lead to worse math performance because of the negative emotions that arise in math-related situations. Furthermore, there was no scientific evidence to support the idea that "men are smarter in mathematics than women"; research continuously shows that there was no apparent gender gap in mathematical abilities. Any discrepancies that do occur are mostly due to cultural and societal factors rather than biological variations.

CHAPTER III

METHODOLOGY

This chapter discusses the methods used by the researchers, the Research Design, the Research Locale, the Respondents, the Samples and Sampling Techniques, the Research Instrument and Validation of Instrument, the Data Gathering Procedure, and Statistical Treatment to conduct the study.

Research Design

This study was a descriptive research on mathematics anxiety in Grade 10 students, aimed to identify the characteristics and prevalence of math anxiety within this demographic. It explored the levels of anxiety experienced, potential contributing factors, and the impact on their academic performance in mathematics. This research did not seek to establish causal relationships between various factors; instead, it focuses on describing the current state of math anxiety among these students by collecting information about their experiences and perceptions related to math.

Research Locale

The location of the study was conducted at Barangay Tomas Cabili, Iligan City. Furthermore, the researchers could easily be available and accessible.

The Respondents

The 30 Grade 10 learners of Tomas Cabili National High School during the school year 2024-2025 were the respondent of the study. All of the participants were selected through random sampling.

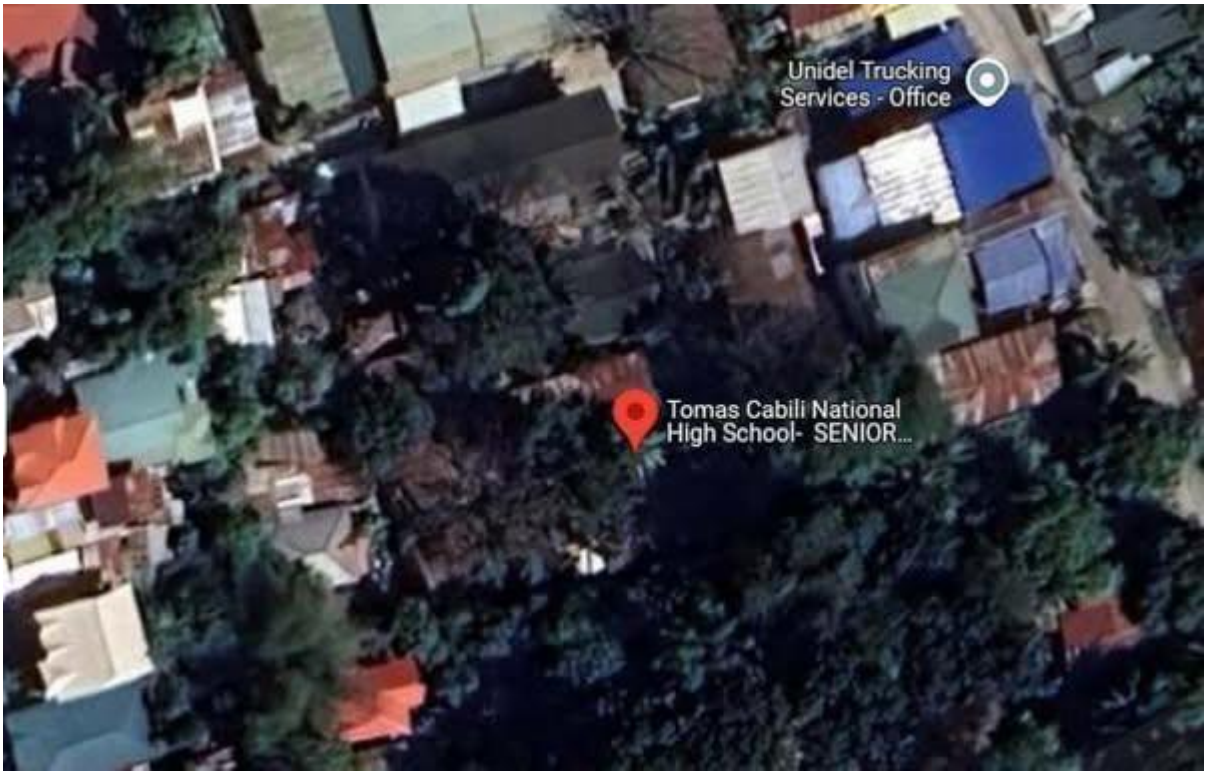


Figure 2. Map of Tomas Cabili National High School

Samples and Sampling Techniques

All of the participants were selected through random sampling. This sampling method was conducted where each member of the population had an equal opportunity to become part of the sample. For this purpose, a survey questionnaire in math anxiety was given to the respondent to answer.

Research Instrument and Validation of Instrument

For this research, a survey questionnaire instrument was used to achieve the main objective of the study. The survey questionnaires were distributed to the selected students. The questionnaire given in the learners aim to assess the level of mathematics anxiety. The questionnaire was structured in a way that respondents would answer it easily. Thus, the set of questionnaire was designed used the Mathematics Anxiety Rating Scale. MARS a rating scale that requires the respondent to indicate his or her degree of anxiety to the statement. In order to test the validity of the evaluation tool which was used for the study researcher assess the answers from the questionnaire given to the 30 respondents.

Data Gathering Procedure

Relevant studies, literature, and other citations were collected from reliable sources. A letter of request was submitted to the principal of Tomas Cabili National High School, seeking permission for the researchers to conduct the study among Grade 10 Acacia students. Once approval was granted, the research implementation commenced. In any research study, proper data gathering and procedures are essential to achieve the desired results. In this study, the researchers employed a survey method, distributing questionnaires to randomly selected respondents, who were all Grade 10 learners. The scores obtained were used to assess their level of

anxiety in mathematics. Additionally, the researchers collected scores from Grade 10 learners using the MARS tool to determine whether they were categorized as Not at all Anxious, Mildly Anxious, Moderately Anxious, Severely Anxious, or Extremely Severely Anxious.

Statistical Treatment

Frequency Count- refers to the number of times a specific value or category appears within a data set, essentially counting how often a particular characteristic occurs in your research sample. Thus, a basic descriptive statistic often presented in a table or graph to visualize the distribution of data across different categories.

Percentage- is a number or ratio express as a fraction of 100. It is often denoted using percentage sign (%).

$P=F/N\times100$

Where in:

P=Percentage

F=Frequency

N=Total number of a respondents

Grading System- is a method for evaluating and categorizing student performance in a class.

Grading Scale	Descriptors	Remarks
90-100	Outstanding	Passed
85-89	Very Satisfactory	Passed
80-84	Satisfactory	Passed
75-79	Fairly Satisfactory	Passed
Below 75	Not Meet Expectations	Failed

Mathematics Anxiety Rating Scale- refers to a survey designed to assess math anxiety. In this questionnaire, individuals evaluate their anxiety levels concerning various math-related situations using a Likert scale, which typically consists of 5 points ranging from "not at all anxious" to "extremely anxious." Scale used for Mathematics Anxiety Rating Scale scores: Range Verbal Description 1Not at all Anxious 2Mild Anxious 3Moderate Anxious 4Severe Anxious 5 Extremely Severe Anxious

Range	Verbal Description
1	Not at all Anxious
2	Mild Anxious
3	Moderate Anxious
4	Severe Anxious
5	Extremely Severe Anxious

CHAPTER IV

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

This chapter presented the analysis and interpretation of the result of Mathematics Anxiety: Its Impact on Tomas Cabili National High School Grade 10-Acacian Students Performance and Well-being.

Presentation of Results

The data for this research were obtained by counting the number of persons upon their response of Mathematics Anxiety: Its Impact on Tomas Cabili National High School Grade 10-Acacian Students Performance and Well-being. The results were summarized and presented in every table.

Table 1. Factors contributing to mathematics anxiety that impact students performance

Factors contributing to mathematics anxiety	Frequency	Percentage (%)
Fear of failure	10	33.3%
Time pressuring during exams	5	16.7%
Difficulty in understanding concepts	11	36.7%
Lack of confidence in abilities	4	13.3%
Negative past experiences	0	0%
Teacher's teaching method	0	0%
Pressure from parents or peers	0	0%
Total	**30**	**100%**

Table 1 showed the result of response of the respondents on factors contributing to mathematics anxiety that impact students performance. Out of thirty (30) respondents, ten (10) or 33.3% answered fear of failure, five (5) or 16.7% answered time pressuring during exams, eleven (11) or 36.7% answered difficulty in understanding concepts, four (4) or 13.3% answered lack of confidence in abilities and 0% answered for negative past experiences, teacher's teaching method, and pressure from parents or peers. This implies that there was difficulty in understanding Mathematics concepts.

Table 2. Anxiety level of the Grade 10-Acacian students of Tomas Cabili National High School.

Levels of Math Anxiety	Frequency	Percentage (%)
Not at all Anxious	5	16.7%
Mild Anxious	6	20%
Moderate Anxious	15	50%
Severe Anxious	1	3.3%
Extremely Severe Anxious	3	10%
Total	**30**	**100%**

Table 2 showed the result of response of the respondents on Anxiety level of the Grade 10-Acacia students of Tomas Cabili National High School. Out of thirty (30) respondents, five (5) or 16.7% answered Not at all Anxious, six (6) or 20% answered Mild Anxious, fifteen (15) or 50% answered Moderate Anxious, one (1) or 3.3% Severe Anxious, three (3) or 10% answered Extremely Severe Anxious. This implies that the Math anxiety level of the Grade 10-Acacian students were Moderate Anxious.

Table 3. Mathematics anxiety affects students academic performance in the subject.

Levels of Math Anxiety	Frequency	Percentage (%)
Not at all Anxious	2	6.7%
Mild Anxious	7	23.3%
Moderate Anxious	16	53.3%
Severe Anxious	4	13.3%
Extremely Severe Anxious	1	3.3%
Total	**30**	**100%**

Table 3 showed the result of response of the respondents on Mathematics anxiety affects students academic performance in the subject. Out of thirty (30) respondents, two (2) or 6.7% answered Not at all Anxious, seven (7) or 23.3% answered Mild Anxious, sixteen (16) or 53.3% answered Moderate Anxious, four (4) or 13.3% Severe Anxious, one (1) or 3.3% answered Extremely Severe Anxious. This implies that the Mathematics anxiety affects students academic performance in the subject was Moderate Anxious.

Table 4. Strategies used to manage mathematics anxiety

Strategies	Frequency	Percentage (%)
Practicing more problems	16	53.3%
Use Ai apps in solving math	2	6.7%
Seeking help from teachers or peers	11	36.7%
Using relaxation techniques (e.x., deep breathing)	1	3.3%
Positive self-talk and motivation	0	0%
Avoiding math-related tasks	0	0%
Avoiding math-related tasks	0	0%
Total	**30**	**100%**

Table 4 showed the result of response of the respondents on strategies used to manage mathematics anxiety. Out of thirty (30) respondents, sixteen (16) or 53.3% answered practicing more problems, two (2) or 6.7% answered use Ai apps in solving math, eleven (11) or 36.7% answered seeking help from teachers or peers, one (1) or 3.3% answered using relaxation techniques (e.x., deep breathing), zero 0 or 0% answered positive self-talk and motivation and avoiding math-related tasks. This implies that practicing more problems were the strategies used to manage mathematics anxiety of the students.

Table 5. Current mathematics grade range of Grade 10-Acacian students.

Grading Scale	Frequency	Percentage (%)
90-100 Outstanding	9	30%
85-89 Very Satisfactory	17	56.7%
80-84 Satisfactory	4	13.3%
75-79 Fairly Satisfactory	0	0%
Below 75 Did Not Meet Expectations	0	0%
Total	**30**	**100%**

Table 5 showed the result of response of the respondents on current mathematics grade range of Grade 10-Acacian students. Out of thirty (30) respondents, nine (9) or 30% answered Outstanding, seventeen (17) or 56.7% answered Very Satisfactory, four (4) or 13.3% answered Satisfactory, zero (0) or 0% answered Fairly Satisfactory and Did Not Meet Expectations. This implies that the current mathematics grade range of Grade 10-Acacian students were 85-89%.

CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATIONS

This chapter contains the summary, conclusion and recommendation of the study.

Summary

The primary purpose of the research was to determine the Mathematics Anxiety: It's Impact on Tomas Cabili National High School Grade 10-Acacia Student's Performance and Well-being. The purpose of the research was to present information for reflecting and planning how mathematics anxiety can be reduced and regulated in classrooms. Additionally, this study seeks to provide valuable insights for educators to developed effective strategies to support students struggling with mathematics anxieties.

The respondents of the study were 30 Grade 10 students from Tomas Cabili National High School, Iligan City.

The researcher used descriptive research. This design was used because it focuses on describing the current state of mathematics anxiety among students by collecting information about their experiences and perception related to math.

Questionnaire was used to gather the data. The Mathematics Anxiety Rating Scale was used in the questionnaire were Extremely Severe Anxious, Severe Anxious, Moderate Anxious, Mild Anxious, Not at all Anxious to determine the mathematics anxiety of the Grade 10-Acacia students.

Based on the results of the study, it found that a significant factor contributing to mathematics anxiety was the difficulty students face in understanding math concepts.

Based on the results of the study, the Mathematics Anxiety level of the Grade 10-Acacian students were Moderate Anxious.

Based on the results of the study, the mathematical anxiety affects students academic performance in the subject was Moderate Anxious.

Based on the results of the study, practicing more problems were the strategies used to manage mathematics anxiety of the students. Thus, helps reinforce conceptual understanding and build confidence.

Lastly, based on the results of the study, their performance remains strong, with grades consistently falling within the very satisfactory range of 85–89%.

Conclusion

The study concludes that while Grade 10-Acacian students experience moderate mathematics anxiety, mainly due to difficulties in understanding math concepts. Furthermore, it does not significantly impact their academic performance. Practicing more problems had been shown to help reduce anxiety and support learning. These findings indicate that further efforts to improve conceptual clarity and encourage regular practice could further alleviate anxiety, leading to better well-being and sustained academic success.

Recommendations

Based from the results and findings presented in the study, the following conditions were drawn:

A. For the students, continue seeking help from teachers and peers. Form study groups, attend tutoring sessions, and utilize supplementary learning resources to build a deeper conceptual understanding of math.

B. For the teachers, incorporate varied teaching approaches (ex., visual aids, hands-on activities, and technology-based tools) to cater to different learning styles and simplify complex math concepts. Foster an atmosphere where students feel safe to express their difficulties without fear of judgment.

C. For the School Administrators, allocate funds for professional development that addresses both effective math instruction and student mental health. Provide access to tutoring programs and learning aids.

D. For the Parents, encourage the students to view math as a learnable skill. Emphasize the value of persistence and effort.

D. For the Future Respondents, investigate the effectiveness of digital learning tools and online resources in reducing math anxiety among students.

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APPENDICES
Appendix A

Letter to the School Principal

Republic of the Philippines
Department of Education
Region X-Northern Mindanao
Division of Iligan City
West 1 District
TOMAS CABILI NATIONAL HIGH SCHOOL
Tomas Cabili, Iligan City

February 25, 2025

To :**MR. BERNIE F. LUBGUBAN**
School Principal
Tomas Cabili National High School
Sir:

Greetings!

We, Grade 10 students from Acacia, are conducting a research study entitled: **Mathematics Anxiety: Its Impact on Tomas Cabili National High School Grade 10 Student's Performance and Well-being** as part of the requirements for the English subject. In this regard, we would like to request for your permission to conduct a questionnaire on Grade 10-Acacia students of Tomas Cabili National High School. Rest assured that a gathered data/information will be protected and will be solely for research purposes only. Your support and kindness on this endeavor is highly appreciated.

Thank you and God bless us all!

Respectfully,

Feliscent Shayne A. Antido

Jowina D. Mainit

Russee R. Manait

Noted: Approved:

MICHAEL VIR D. DE GUZMAN **BERNIE F. LUBGUBAN**
English Teacher Principal

Appendix B

Letter to the Respondents

Tomas Cabili, Iligan City

February 25, 2025

Dear Respondents,

Greetings!

We, the Grade 10 researchers conducted a research study entitled “Mathematics Anxiety: Its Impact on Tomas Cabili National High School Grade 10-Acacian Student's Academic Performance and Well-being.” We, the researchers asking for your permission and valuable time to become one of our respected respondents of this study.

Hoping for your cooperation and positive response, God bless us all!

Respectfully yours,

Feliscent Shayne A. Antido

Jowina D. Mainit

Russee R. Manait

Noted by:

_____

Michael Vir D. De Guzman

English Subject Teacher

Appendix C

Documentation



Figure 3. Documentation

Appendix D
Survey Questionnaire

PART I. STUDENT INFORMATION
Direction: Please answer the question below by writing the one that best describes you.

I. Profile

Name
(Optional)___

Directions: Please put a check/number on the space provided for your answer.

1. Age ___ years old
2. Sex ____Male
   ____Female
3. Monthly family income____

PART II: Mathematics Anxiety: Its Impact on Tomas Cabili National High School Grade10-Acacia Student's Performance and Well-being.
Directions: Please put a check on the box provided for your answer.

1. What are the factors contributing to mathematics anxiety that impact students performance? (Choose only one answer)
   - ☐ Fear of failure
   - ☐ Time pressure during exams
   - ☐ Difficulty in understanding concepts
   - ☐ Lack of confidence in abilities
   - ☐ Negative past experiences
   - ☐ Teacher’s teaching method
   - ☐ Pressure from parents or peers
   - ☐ Other (Please Specify):

2. What is the anxiety level of Grade 10-Acacia students of Tomas Cabili National High School?

   ☐1 Not at all Anxious

   ☐2 Mild Anxious

   ☐3 Moderate Anxious

   ☐4 Severe Anxious

   ☐5Extremely Severe Anxious

3. How does mathematic anxiety affect students academic performance in the subject?

   ☐1 Not at all Anxious

   ☐2 Mild Anxious

   ☐3 Moderate Anxious

   ☐4 Severe Anxious

   ☐5 Extremely Severe Anxious

4. Which of the following strategies do you use to manage your mathematics anxiety? (Choose only one answer)

   ☐ Practicing more problems

   ☐ Use AI apps in solving math

   ☐ Seeking help from teachers or peers

   ☐ Using relaxation techniques (e.g., deep breathing)

   ☐ Positive self-talk and motivation

   ☐ Avoiding math-related tasks

   ☐ Other (Please Specify):_____

5. What is the current mathematics grade range of Grade 10-Acacia students?

☐ 90-100 Outstanding

☐ 85-89 Very Satisfactory (Passed)

☐ 80-84 Satisfactory (Passed)

☐ 75-79 Fairly Satisfactory (Passed)

☐ Below 75 Did Not Meet Expectations (Failed)

Survey Questionnaire

PART I. STUDENT INFORMATION
Direction: Please answer the question below by writing the one that best describes you.

I. Profile

Name
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☐ Below 75 Did Not Meet Expectations (Failed)

CURRICULUM VITAE

I. PERSONAL INFORMATION



- Name : Russee R. Manait
- Date of Birth : February 24, 2009
- Place of Birth : E&R Hospital, Iligan City, Philippines
- Nationality : Filipino
- Languages Spoken : Cebuano, English and Tagalog

II. EDUCATIONAL BACKGROUND

Level	School Name	Year Attended
Grade 1-6	Rufino Mangilisan Memorial Elementary School	2014-2015
Grade 7-9	Digkilaan National High School	2021-2024
Grade 10	Tomas Cabili National High School	2024-2025

III. SKILLS & ABILITIES

- Public Speaking
- Dancer
- Artist

IV. CHARACTER REFERENCES

Michael Vir D. De Guzman
English Department Head -Tomas Cabili National High School
09530429215

Angela P. Canoy
Teacher II -Tomas Cabili National High School
099533211134

Marie Christine Pastor
Teacher -Tomas Cabili National High School
09453782022

CURRICULUM VITAE



I. PERSONAL INFORMATION

- Name : Jowina D. Mainit
- Date of Birth : September 01, 2009
- Place of Birth : Iligan City, Philippines
- Nationality : Filipino
- Languages Spoken : Filipino, Cebuano, and English

II. EDUCATIONAL BACKGROUND

Level	School Name	Year Attended
Grade 7-10	**Tomas Cabili National High School**	**2021-2025**
Grade 1-6	**Tomas Cabili Central School**	**2014-2020**

III. SKILLS & ABILITIES

- Crocheting
- Drawing

IV. CHARACTER REFERENCES

Michael Vir D. De Guzman
English Department Head -Tomas Cabili National High School
09530429215

Angela P. Canoy
Teacher II -Tomas Cabili National High School
09533211134

Marie Christine T. Pastor
Teacher -Tomas Cabili National High School
09453782022

CURRICULUM VITAE



I. PERSONAL INFORMATION

- Name : Feliscent Shayne A. Antido
- Date of Birth : May 10, 2009
- Place of Birth : Iligan City, Philippines
- Nationality : Filipino
- Languages Spoken : Filipino, Cebuano, and English

II. EDUCATIONAL BACKGROUND

Level	School Name	Year Attended
Grade 7-10	**Tomas Cabili National High School**	**2021-2025**
Grade 1-6	**Tomas Cabili Central School**	**2014-2020**

III. SKILLS & ABILITIES

- Video Editing
- Sketching
- Writing

IV. CHARACTER REFERENCES

Michael Vir D. De Guzman
English Department Head -Tomas Cabili National High School
09530429215

Angela P. Canoy
Teacher II -Tomas Cabili National High School
09533211134

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